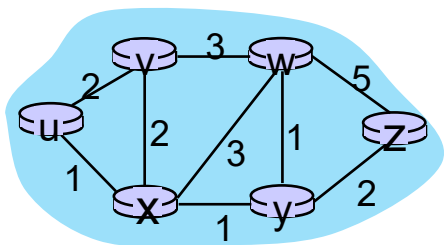


Computer Networks

Intra-AS Routing

RIP and OSPF

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Making Routing Scalable

Routing studied thus far - idealized

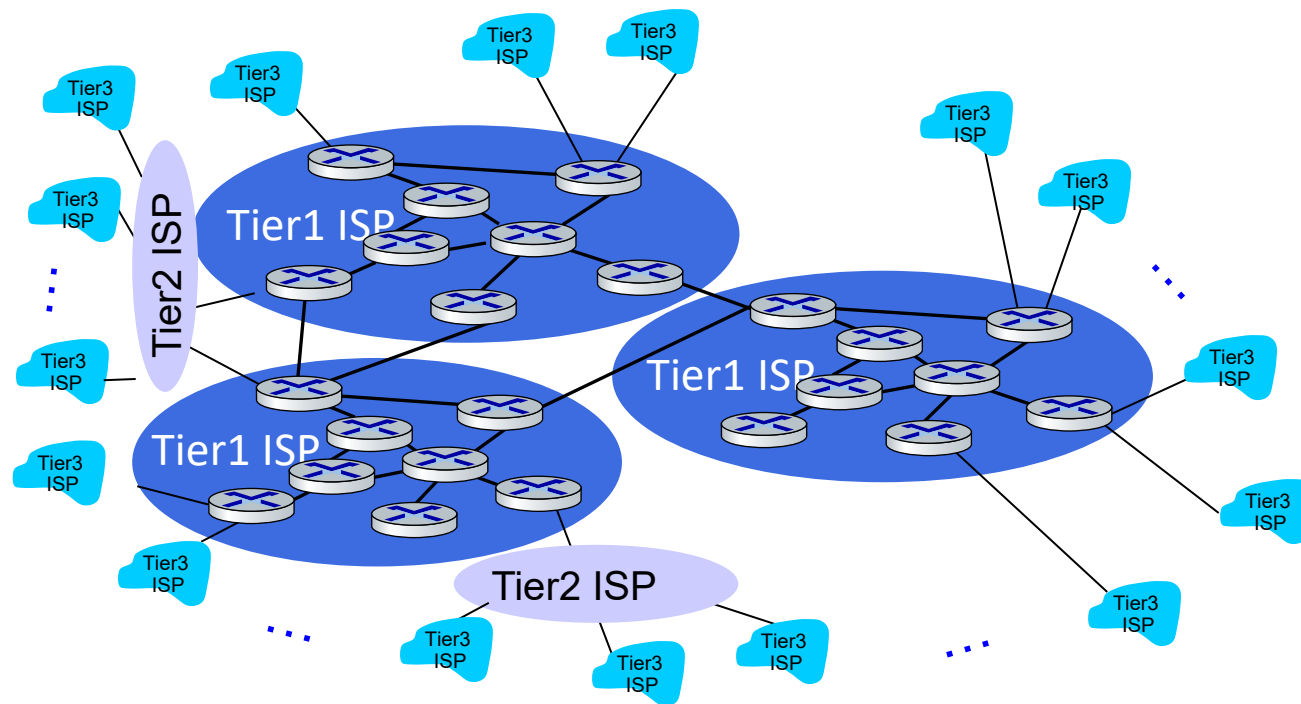
- All routers identical
- Network “flat”
- This is not true in practice

Scale: billions of destinations:

- Can't store all destinations in routing tables
- Routing table exchange would swamp links

Administrative autonomy:

- Internet: a network of networks
- Each network admin may want to control routing in its own network



Internet Approach to Scalable Routing

Autonomous systems (AS) (a.k.a. “**domains**”): group of routers controlled by a single administrative entity, i.e. an ISP, a company, a university

Intra-AS (aka “intra-domain”):

routing within an AS (“network”)

- All routers in AS must run same intra-domain protocol
- Routers in different AS can run different intra-domain routing protocols
- **Gateway router**: at “edge” of its own AS, has link(s) to router(s) in other AS'es
- Also called **Interior Gateway Protocol**

Inter-AS (aka “inter-domain”):

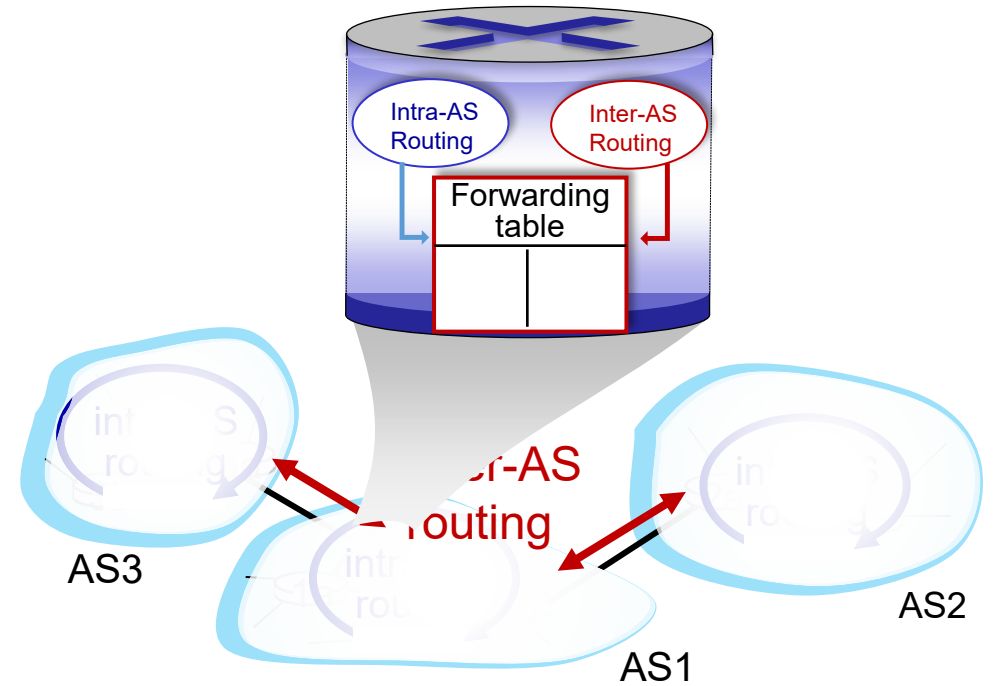
routing **among** AS'es

- Gateways perform inter-domain routing (as well as intra-domain routing)
 - Also called **Exterior Gateway Protocol**
-

Interconnected ASes

Forwarding table configured by intra- and inter-AS routing algorithms

- Intra-AS routing determine entries for destinations within AS
- Inter-AS & intra-AS determine entries for external destinations

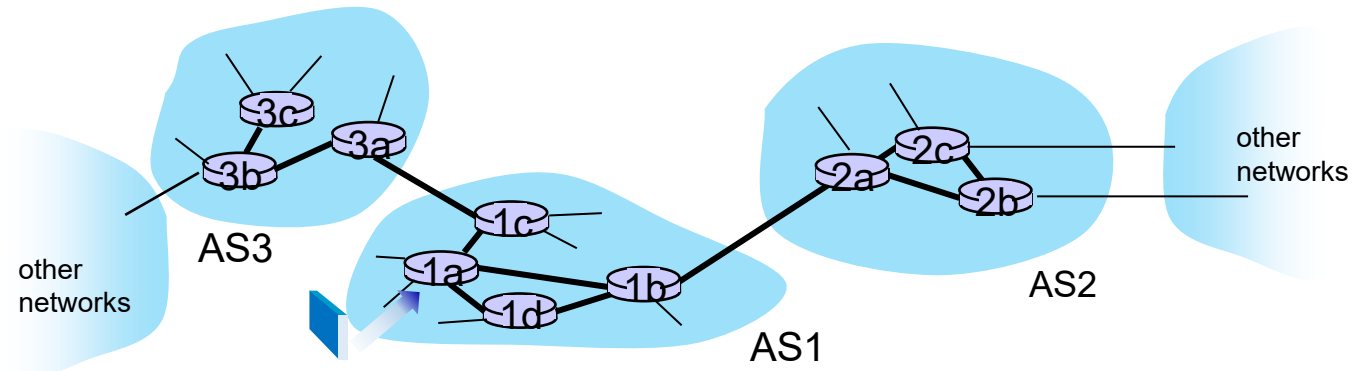


Inter-AS Routing: A Role in Intra-domain Forwarding

- Suppose router in AS1 receives datagram destined outside of AS1:
 - Router should forward packet to gateway router in AS1, but which one?

AS1 inter-domain routing must:

1. Learn which destinations reachable through AS2, which through AS3
2. Propagate this reachability info to all routers in AS1



Intra-AS Routing: Routing within an AS

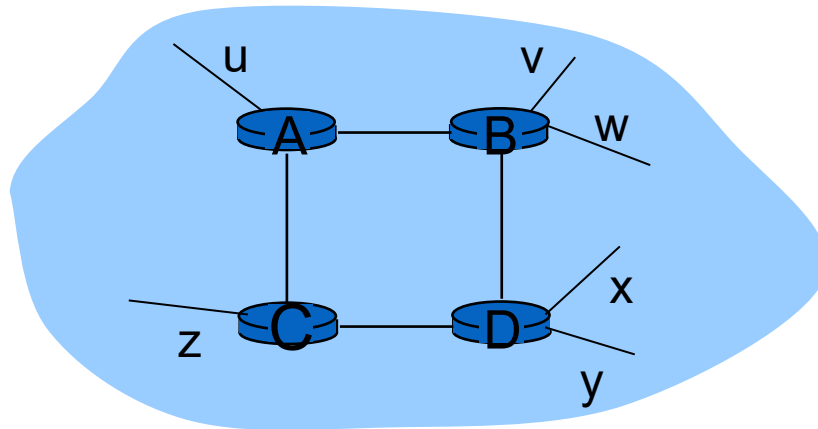
Most common intra-AS routing protocols:

- **RIP: Routing Information Protocol** [RFC 1723]
 - Classic DV
 - No longer widely used
 - **EIGRP: Enhanced Interior Gateway Routing Protocol**
 - DV based
 - Formerly Cisco-proprietary for decades (became open in 2013 [RFC 7868])
 - **OSPF: Open Shortest Path First** [RFC 2328]
 - Link-state routing
 - **IS-IS** protocol (ISO standard, not RFC standard) essentially same as OSPF
-

Routing Information Protocol

RIP (Routing Information Protocol)

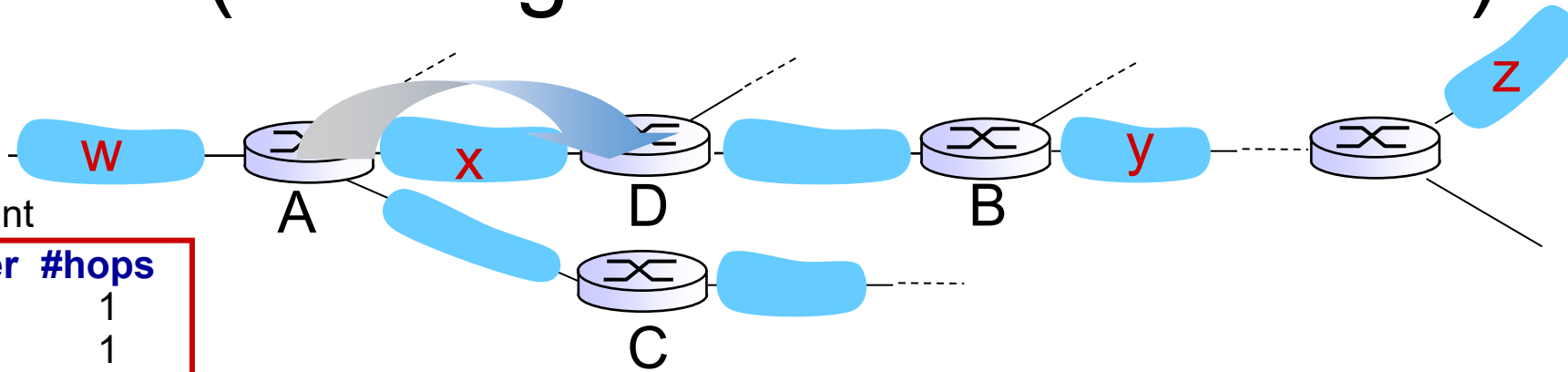
- Included in BSD-UNIX distribution in 1982
- Distance vector algorithm
 - Distance metric: # hops (max = 15 hops), each link has cost 1
 - Hop: number of subnets traversed along the shortest path from source router to destination subnet
 - DVs exchanged with neighbors every 30 sec in response message (aka advertisement)
 - Each advertisement: list of up to 25 destination subnets (in IP addressing sense)



From router A to destination subnets:

<u>Subnet</u>	<u>Hops</u>
u	1
v	2
w	2
x	3
y	3
z	2

RIP (Routing Information Protocol)



A-to-D advertisement

Dest	Next-router	#hops
W	-	1
X	-	1
Z	C	4

Routing table in router D

Destination subnet	Next-router	# hops to dest
W	A	2
y	B	2
Z	B A	7 5
X	--	1
....		

RIP: Link Failure and Recovery

If no advertisement heard after 180 sec → neighbor/link declared dead

- Routes via neighbor invalidated
 - New advertisements sent to neighbors
 - Neighbors in turn send out new advertisements (if tables changed)
 - Link failure info quickly propagates to entire net
 - **Poisoned reverse** used to prevent ping-pong loops (infinite distance = 16 hops)
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OSPF (Open Shortest Path First) Routing

OSPF (Open Shortest Path First) Routing

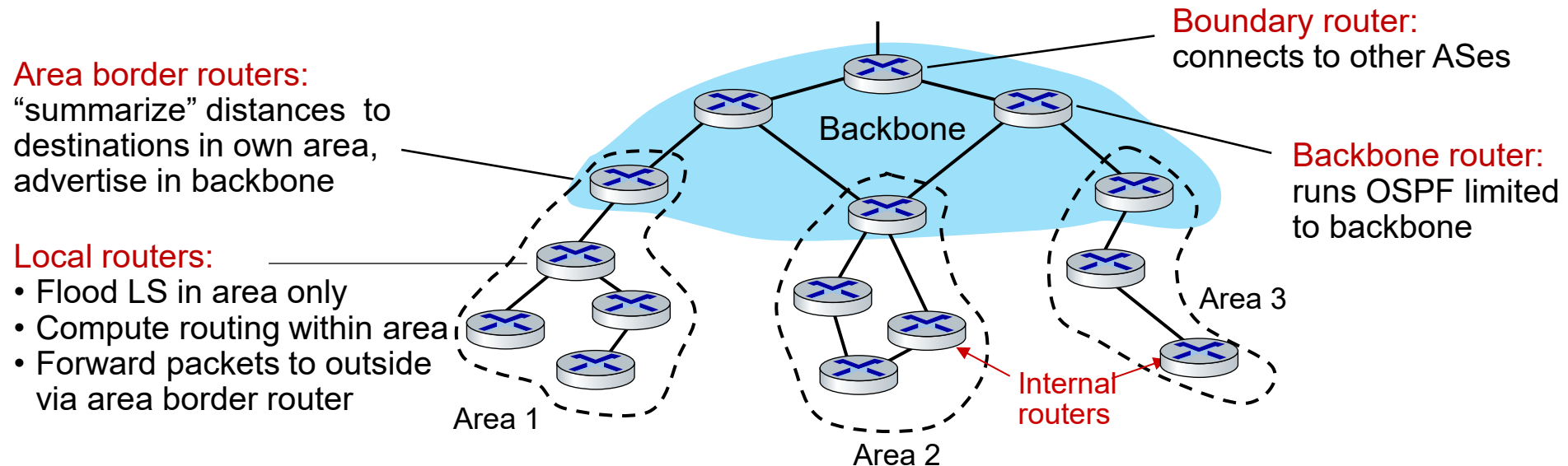
- “Open”: Publicly available
 - Classic link-state
 - Each router floods OSPF link-state advertisements to all other routers in entire AS
 - Multiple link costs metrics possible: bandwidth, delay
 - Each router has full topology, uses Dijkstra’s algorithm to compute forwarding table
-

OSPF : Advanced Features

- **Security:** All OSPF messages can be authenticated (to prevent malicious intrusion)
 - **Multiple same-cost paths** allowed (only one path in RIP)
 - For each link, multiple cost metrics for different **ToS**
 - Satellite link cost set low for best effort ToS; high for real-time ToS
 - Integrated unicast and **multicast** support:
 - Multicast OSPF (MOSPF) uses same topology data base as OSPF
 - **Hierarchical** OSPF in large domains
-

Hierarchical OSPF

- **Two-level hierarchy:** local area, backbone.
 - Link-state advertisements flooded only in area, or backbone
 - Each node has detailed area topology; only knows direction to reach other destinations



Summary

❑ Internet Approach to Scalable Routing:

- Intra-AS and Inter-AS

❑ Intra-AS routing:

- RIP: Routing Information Protocol
 - OSPF: Open Shortest Path First
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